

Jaygovindhan Neelananarayanan

Game Developer · AI Tooling · Product Engineering

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PROFESSIONAL SUMMARY

Software developer with shipped commercial games experience and a strong foundation in performance-critical systems, game architecture, and mobile development. Two commercial mobile titles shipped to Google Play and App Store. Built production-grade C++17 systems including physics engines, FSM AI, real-time rendering pipelines, and data-driven game architectures across five academic projects. Experienced in cross-functional teams, monetisation design, and iterative live-game development. Active daily user of AI coding tools (Claude Code, Copilot) as part of engineering workflow. Currently completing an MSc in Computer Games Technology with VR at City St George's, University of London (2024-2026).

KEY SKILLS

Languages: C/C++ (C++17), C#, Python, JavaScript, TypeScript

Game Development: Unity 6 LTS, Unreal Engine 4/5, MonoGame/XNA, OpenGL/GLSL — shipped production experience across mobile and desktop

Systems & Architecture: FSM design, event-driven systems, observer pattern, object pooling, resource management, data-driven JSON config, separation of concerns

Mobile & Performance: Android and iOS game delivery, frame-rate optimisation, GPU profiling, performance budget management, real-time rendering

Monetisation & Live Ops: IAP system design, monetisation mechanics, offer and progression systems, A/B test-informed iteration, player behaviour research

Testing & Quality: Unit and integration testing, code review, controlled experiment design, quantitative evaluation

AI & Generative AI: Active daily use of Claude Code and Copilot for code generation, debugging, and review; applied Generative AI tooling in production workflow

Tools: Git / GitHub, Visual Studio, Bullet3, FMOD Core, Assimp, Autodesk Maya, ImGui, ScriptableObject config

Languages Spoken: English (Native), Tamil (Native), Hindi (Limited Working), Arabic (Elementary), French (Elementary), Portuguese (Elementary)

WORK EXPERIENCE

Computer Games Designer (Freelance, Hourly) | ALPHA2 Solutions & Software Limited

Jul 2026 - Present | London

- Engaged on an hourly, as-required basis, contributing game design work to a client project delivered through ALPHA2 Solutions & Software.

Developer & Technical Artist | OFGA Multimedia

Nov 2022 - Aug 2024 | Remote

- Shipped a commercial mobile title end-to-end — contributing to game logic, performance-critical rendering systems, and asset pipelines within a live production codebase serving thousands of active users on Android and iOS.
- Maintained performance budgets and operational stability across the full development lifecycle — development, integration, testing, and post-launch maintenance — within a small, agile cross-functional team.
- Worked across engineering and technical art simultaneously: integrating self-produced assets, coordinating runtime behaviour with programmers, and iterating on visual features under commercial release pressure.

Game Designer | ChennaiGames Studio

Jun 2021 - Oct 2022 | Chennai, India

- Led full design lifecycle on two commercial mobile titles — RADDX Racing Metaverse and Hurley Super Surfer — shipped to Google Play and App Store, owning game design documentation, monetisation mechanics, player behaviour research, and product architecture (including metaverse architecture for RADDX).
- Designed IAP systems, progression-linked offer structures, and engagement loops — presented to production leads and implemented by engineering teams; iterated using live player behaviour data.
- Coordinated cross-functional delivery across programmers, artists, and QA under tight commercial release cycles.

ADDITIONAL WORK EXPERIENCE

Team Member | The Salad Project

Jul 2026 - Present | London

- Point-of-sale operations, food preparation, and customer service in a high-volume fast-casual environment, maintained alongside full-time MSc study.

Sales Assistant | The Salad Kitchen

Mar 2026 - Jun 2026 | London

- Customer service and retail food operations in a fast-paced hospitality environment, balanced alongside MSc dissertation work.

Front of House Staff Member | UBA Shoreditch

Aug 2025 - Nov 2025 | London

- Front-of-house service and customer-facing operations in a high-volume café/bar environment.

PROJECTS

8-Bit Bounce — 3D Platformer in C++/OpenGL | City St George's | INM710 Advanced Games Technology

2025 | London

- Built a self-contained 3D platformer in C++17 and OpenGL — four-state FSM enemy AI (patrol, chase, attack, dying) across 20 concurrent agents, driven by line-of-sight raycasting, proximity, and health state transitions.
- Implemented Bullet3 rigid-body physics covering gravity, restitution, surface friction variation, impulse forces on collision, and per-frame player synchronisation to oscillating platforms.
- Integrated FMOD Core spatialised audio and Assimp mesh loading; full game loop with score/timer/health HUD, power-ups, skybox, wire-frame debug toggle, and alpha fade spawn effects — maintained a stable frame-rate throughout.

Side Scroller Shooter — 2D Arcade Game in C#/MonoGame | City St George's | INM379 Computer Games Architecture

2025 | London

- Production-grade engine architecture: observer/EventManager pattern, IGameState FSM screen flow, wave-based enemy spawning, three-life system, power-ups, and high-score persistence.
- Difficulty system driven entirely by JSON config — zero hardcoded spawn values, enabling rapid iteration on game balance without touching code.
- ResourceManager caches all textures, fonts, and SFX on first load; object pooling and spatial partitioning implemented for cache-locality performance.

Forklift Idle — Logistics Simulation & Idle Game | Personal Project (In Development)

2025-Present

- Building a delivery/logistics idle simulation game: data-driven progression system driven entirely by JSON config, observer-pattern event architecture decoupling game logic from UI state, zero hardcoded values across resource generation, delivery loop, and upgrade tiers.
- Applying monetisation and retention design knowledge from two shipped commercial titles — balancing idle loop engagement curves and designing progression-linked offer systems built for live-service iteration.

Chromacheck — Unity Accessibility Compliance Toolkit | City St George's | INM373 Individual Research Project

2025-Present | London

- Designing a Unity Editor accessibility compliance tool: GPU shader pipeline for colour-blindness simulation (Machado et al. 2009 CVD matrices), automated WCAG 2.1 contrast-ratio scanner, and web-facing JSON reporting interface — currently in active development as MSc dissertation.

Drive Circuit — OpenGL Racing Game | City St George's | INM376 Computer Graphics

2024-2025 | London

- C++/GLSL racing game: Catmull-Rom spline track generation, TNB-frame cameras, multi-texturing, dynamic lighting modes, and a GLSL rain/lightning weather shader.

EDUCATION

MSc Computer Games Technology with VR | City St George's, University of London

2024-2026 (Expected) | London

- Computer Graphics (C++/OpenGL/GLSL), Computer Games Architecture (C#/MonoGame), Advanced Games Technology (C++17/OpenGL, physics, AI), Game Development Process (Unity 6), Virtual Reality Systems, Individual Research Project (in progress).

BSc Media Technology — Game Design Specialisation | ICAT College of Design & Media

2018-2021 | Chennai, India

CERTIFICATIONS

- Microsoft Security Essentials Professional Certificate — Microsoft & LinkedIn (2025) | AI Security, Cloud Security, GRC
- Introduction to AI Governance — LinkedIn Learning
- Ethics in the Age of Generative AI — LinkedIn Learning (Feb 2026)
- Unity 3D 2021 Essential Training — LinkedIn Learning
- Game Design and Development 1: 2D Shooter — Michigan State University (2024) | C#, OOP, Software Architecture